Network Security Investigation

****Tools:** Nmap 7.99.1 and Wireshark**

****Target:** Own Windows computer**

****Project Type:** Practical Cybersecurity Investigation**

1. Introduction

Network security is an important part of protecting computers and the information they handle. Computers use network ports and services to communicate with other devices. If these services are not configured properly, they may create unnecessary security risks.

In this project, I investigated the network services running on my own Windows computer using Nmap. I also used Wireshark to become familiar with capturing network traffic and examining packets.

The purpose was to understand how a basic network investigation is performed and how its findings can be recorded in a clear and organized manner.

2. Objectives

The main objectives of this project were:

- To identify open TCP ports on my Windows computer.

- To understand the services associated with the open ports.

- To learn the basic use of Nmap for service discovery.

- To become familiar with packet capture and analysis using Wireshark.

- To document the findings and understand the limitations of the investigation.

3. Tools and Technologies

Nmap

Nmap is a network scanning tool used to identify active hosts, open ports, and network services. In this investigation, it was used to scan my own computer and attempt to identify the services associated with open TCP ports.

Wireshark

Wireshark is a network protocol analyzer. It captures packets visible on a selected network interface and allows users to examine communication protocols and packet details.

Windows

My own Windows computer was used as the target system for the investigation.

4. Scope and Authorization

The investigation was limited to my own Windows computer on a local network. The purpose was educational and focused on service discovery and network observation.

No exploitation, password attacks, or destructive testing was performed.

5. Methodology

The investigation followed these steps:

1. Identified the local IPv4 address of the Windows computer.
2. Ran Nmap service detection against the computer.
3. Recorded the scan output for documentation.
4. Opened Wireshark and performed a short initial packet capture.
5. Reviewed the Nmap results and documented the findings.

6. Considered how packet analysis could provide additional information about network communication.

The Nmap command used was:

```powershell

nmap -sV <YOUR IP ADDRESS>

```

The address shown above was the local IP address reported for the target during the recorded scan. It should be replaced in any future reproduction if the computer receives a different address.

6. Nmap Scan Results

The scan was completed successfully. Nmap reported that the target host was up and identified three open TCP ports.

| Port | State | Service reported | Description |

|---|---|---|---|

| 135/tcp | Open | Microsoft RPC | Used by Windows for communication between system components and services |

| 139/tcp | Open | NetBIOS-SSN | Associated with legacy Windows networking |

| 445/tcp | Open | microsoft-ds? | Commonly associated with Windows SMB networking and file sharing |

The scan output also reported 997 closed TCP ports that were not displayed individually. The scan took approximately 9.59 seconds.

Nmap identified the operating system family as Windows. The service identification for port 445 included a question mark, meaning the reported service identification was not fully conclusive.

6.1 Port 135 — Microsoft RPC

Microsoft Remote Procedure Call (RPC) supports communication between Windows components and services. This port is commonly present on Windows systems.

Its presence alone does not indicate that the system is vulnerable. The network exposure and configuration of the related services should be considered.

6.2 Port 139 — NetBIOS Session Service

NetBIOS Session Service supports legacy Windows network communication and may be associated with older file-sharing mechanisms.

If this service is not required, its exposure can be reviewed through Windows networking settings and firewall rules.

6.3 Port 445 — Windows SMB-Related Service

TCP port 445 is commonly associated with Server Message Block (SMB), which supports Windows file and printer sharing and related network functions.

SMB should be configured carefully, particularly on untrusted networks. Access should be restricted to devices and networks that genuinely require it.

The scan did not prove that file sharing was accessible to an unauthorized person or that a vulnerability existed.

7. Wireshark Capture

Wireshark was opened on the active network interface, and a short initial capture was made.

The capture was intended to provide practical experience with packet collection and protocol filtering. However, detailed packet-level observations from this initial capture were not recorded in the available evidence.

For this reason, no specific packet counts, DNS responses, SMB sessions, or ICMP observations are claimed in this report.

Further analysis can be performed by capturing traffic while generating a DNS lookup using:

```powershell
nslookup example.com
```

The following Wireshark display filters can then be used:

- **`dns` to examine DNS packets.**
- **`tcp` to examine TCP communication.**
- **`smb || smb2` to check for SMB traffic.**
- **`icmp` to check for ICMP packets.**

The results should be recorded according to what is actually visible in the capture. A filter returning no packets during a short capture is also a possible observation.

8. Correlation of Findings

Nmap and Wireshark serve different purposes in a network investigation.

Nmap identified ports that appeared to be open on the Windows computer. Wireshark, in contrast, can show packets that pass through the selected network interface during a capture.

The Nmap results establish that ports 135, 139, and 445 were reported as open during the scan. The available Wireshark information is not sufficient to confirm whether packets related to these services appeared in the initial capture.

The tools can provide complementary evidence, but their results must be interpreted separately before drawing conclusions.

9. Security Assessment

The scan identified three open TCP ports associated with common Windows networking functions.

These results suggest areas that can be reviewed as part of routine system hardening. However, an open port does not automatically represent a security weakness.

A more complete assessment would require checking the relevant Windows services, firewall rules, update status, sharing configuration, and network exposure.

No confirmed vulnerability was established by the evidence collected for this project.

10. Recommendations

Based on the initial results, the following general recommendations are appropriate:

1. Keep Windows and installed software updated.

2. Review Windows Firewall rules for RPC, NetBIOS, and SMB-related traffic.

3. Disable network services that are not needed, after checking their dependencies.

4. Restrict file-sharing access to trusted devices and networks.

5. Avoid exposing SMB-related services directly to the public internet.

6. Perform additional packet analysis and document the actual observations.

7. Repeat the scan after relevant configuration changes and compare the results.

These recommendations are general hardening measures and do not mean that a vulnerability was found.

11. Limitations

This project has several limitations:

- The Nmap scan examined the common TCP ports scanned by default, rather than every possible port and protocol.

- Service and version identification can be incomplete.

- The Wireshark capture was short.

- Detailed packet-level findings from the initial capture were not documented.

- No exploitation or comprehensive vulnerability testing was performed.

Because of these limitations, the results should be treated as an introductory investigation rather than a complete security audit.

12. Conclusion

This project provided practical experience with Nmap and Wireshark as basic network investigation tools.

Nmap identified three open TCP ports on my Windows computer: 135, 139, and 445. Reviewing these results helped me understand the role of common Windows networking services and why unnecessary network exposure should be reviewed.

The project also introduced the process of capturing traffic in Wireshark. Detailed packet findings still need to be recorded from the actual capture before stronger conclusions can be made.

Overall, the investigation helped me understand the difference between identifying open services and proving a security vulnerability. It also highlighted the importance of collecting evidence, documenting limitations, and making conclusions based on observed results.

Appendix A: Recorded Scan Summary

- **Nmap version:** 7.99.1
- **Target:** Own Windows computer
- **Host status:** Up
- **Open TCP ports:** 135, 139, 445
- **Operating system family reported:** Windows
- **Scan duration:** Approximately 9.59 seconds

Appendix B: Evidence Checklist

The following evidence can be maintained with the project:

- Nmap scan output saved as `evidence/nmap-scan.txt`
- Wireshark capture saved as `evidence/wireshark-capture.pcapng`
- Investigation notes saved as `analysis/findings.md`
- Final report saved in the `report` folder

Before making the repository public, review the capture and screenshots for private IP addresses, DNS requests, device details, and other sensitive information.

End of Report